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Voltage regulator circuit for a switching circuit load

Abstract

A voltage regulator receives a reference voltage and generates a regulated voltage using a MOSFET having a gate terminal configured to receive a control voltage. A charge pump receives the regulated voltage and generates a charge pump voltage in response to an enable signal and a clock signal generated in response to the enable signal. The voltage regulator further includes a first switched capacitor circuit coupled to the gate terminal and configured to selectively charge a first capacitor with a first current and impose a first voltage drop on the control voltage in response to assertion of the enable signal. The voltage regulator also includes a second switched capacitor circuit coupled to the gate terminal and configured to selectively charge a second capacitor with a second current and impose a second voltage drop on the control voltage in response to one logic state of the clock signal.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application for patent Ser. No. 17/582,431, filed Jan. 24, 2022, the disclosure of which is incorporated by reference.

TECHNICAL FIELD

(1) The present invention generally relates to a voltage regulator circuit and, in particular, to a low drop out (LDO) voltage regulator circuit.

BACKGROUND

(2) FIG. 1 shows a circuit diagram for a conventional voltage regulator circuit **10** of the low drop out (LDO) type. The circuit includes a differential amplifier circuit **12** (functioning as an error amplifier), a metal oxide semiconductor field effect transistor (MOSFET) device **14** and a feedback circuit **16**. An inverting (−) input of the differential amplifier circuit **12** is coupled to receive a reference voltage V_{ref} generated, for example, by a bandgap voltage generator circuit (i.e., $V_{ref}=V_{bg}$). A non-inverting (+) input of the differential amplifier circuit **12** is coupled to receive a feedback voltage V_{fb} output by the feedback circuit **16**. The differential amplifier circuit **12** is powered from a power supply node V_{dd} referenced to a ground node and generates a control voltage V_{ctrl} dependent on an amplified difference between the voltages V_{ref} and V_{fb} . The MOSFET device **14** is a p-channel type device (pMOS transistor) having a source terminal coupled, preferably connected, to the power supply node to receive the supply voltage V_{dd} , a gate terminal coupled, preferably connected, to an output of the differential amplifier circuit **12** to receive the control voltage V_{ctrl} , and a drain coupled, preferably connected, to an output node **18** where a regulated output voltage V_{reg} is generated and applied to a load circuit **20**. The feedback circuit **16** is formed by a resistive voltage divider circuit including a first resistor R_1 coupled, preferably connected, in series at a feedback node **22** to a second resistor R_2 . The first resistor R_1 is coupled, preferably connected, to the output node **18** and the second resistor R_2 is coupled, preferably connected, to the ground node. The feedback voltage V_{fb} is generated at the feedback node **22** which is coupled, preferably connected, to the non-inverting (+) input of the differential amplifier circuit **12**.

(3) There exist a number of applications where voltage regulator circuit **10** supplies the regulated output voltage V_{reg} to a load circuit **20** which operates in a manner where large amplitude current spikes are sunk to ground. An example of such a load circuit **20** is a charge pump that is operating to convert the regulated output voltage V_{reg} to a pumped output voltage V_{cp} . A large amplitude current is sourced in response to switching operations of a pump capacitor in the charge pump circuit to the output node **18**. Another example of such a load circuit **20** is a switched mode DC-DC converter. The voltage regulator circuit **10** must generate a stable regulated output voltage V_{reg} for the load circuit **20** notwithstanding the changes in the instantaneous transient current needs of the load circuit **20**.

(4) One solution is to install a large capacitance filtering capacitor coupled between the output node **20** and ground to smooth the regulated output voltage V_{reg} . In many integrated circuit applications, however, this solution is not acceptable because of the large amount of integrated circuit area that is occupied by the filtering capacitor.

- (5) Another solution is to bias the output stage of the error amplifier with a larger quiescent current. In many integrated circuit applications, however, this solution is not acceptable because the larger quiescent current corresponds to an undesired increase in power consumption.
- (6) Guaranteeing both a sufficiently low output voltage drop and a small quiescent current presents a significant challenge for designing the voltage regulator circuit for supplying a regulated output voltage to a load circuit which sinks large amplitude current spikes.
- (7) There is a need in the art to address the foregoing problems with conventional voltage regulator circuits.

SUMMARY

- (8) In an embodiment, a voltage generator circuit comprises: a voltage regulator circuit configured to receive a reference voltage and generate a regulated output voltage, said voltage regulator circuit including a metal oxide field effect transistor (MOSFET) device having a gate terminal configured to receive a control voltage and a drain terminal configured to source an output current for generating the regulated output voltage; and a charge pump circuit configured to receive the regulated output voltage and generate a charge pump output voltage, said charge pump circuit controlled for operation by an enable signal and a clock signal, wherein the clock signal is generated in response to the enable signal. The voltage regulator circuit includes: a first switched capacitor circuit coupled to the gate terminal and configured to selectively charge a first capacitor with a first current and impose a first voltage drop on the control voltage in response to assertion of the enable signal; and a second switched capacitor circuit coupled to the gate terminal and configured to selectively charge a second capacitor with a second current and impose a second voltage drop on the control voltage in response to one logic state of the clock signal.
- (9) In an embodiment, a circuit comprises: a voltage regulator circuit configured to receive a reference voltage and generate a regulated output voltage, said voltage regulator circuit including a metal oxide field effect transistor (MOSFET) device having a gate terminal configured to receive a control voltage and a drain terminal configured to source an output current for generating the regulated output voltage; and a load circuit configured to receive the regulated output voltage, said load circuit subject to transient current spiking in response to a load circuit control signal. The voltage regulator circuit further includes a switched capacitor circuit coupled to the gate terminal and configured to selectively charge a capacitor with a charging current and impose a voltage drop on the control voltage in response to assertion of the load circuit control signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a better understanding of the embodiments, reference will now be made by way of example only to the accompanying figures in which:
- (2) FIG. 1 is a circuit diagram of a conventional low drop out (LDO) type voltage regulator circuit driving a load circuit;
- (3) FIG. 2 is a circuit diagram of an embodiment for an LDO type voltage regulator circuit;
- (4) FIG. 3 is a circuit diagram of an alternative embodiment for an LDO type voltage regulator circuit;
- (5) FIG. 4 is a circuit diagram of an alternative embodiment for an LDO type voltage regulator circuit;
- (6) FIG. 5 is block diagram of a voltage supply circuit;
- (7) FIG. 6 which illustrates a memory device; and
- (8) FIG. 7 illustrates an integrated system on chip (SoC) device.

DETAILED DESCRIPTION

- (9) Reference is now made to FIG. 2 which shows a circuit diagram of an embodiment for an LDO

type voltage regulator circuit **40**. The circuit includes a differential amplifier circuit **42** (functioning as an error amplifier), a metal oxide semiconductor field effect transistor (MOSFET) device **44** and a feedback circuit **46**. An inverting ($-$) input of the differential amplifier circuit **42** is coupled to receive a reference voltage V_{ref} generated, for example, by a bandgap voltage generator circuit (i.e., $V_{ref}=V_{bg}$). A non-inverting ($+$) input of the differential amplifier circuit **42** is coupled to receive a feedback voltage V_{fb} output by the feedback circuit **46**. The differential amplifier circuit **42** is powered from a power supply node V_{dd} referenced to a ground node and generates a control voltage V_{ctrl} dependent on an amplified difference between the voltages V_{ref} and V_{fb} . The MOSFET device **44** is a p-channel type device (pMOS transistor) having a source terminal coupled, preferably connected, to the power supply node to receive the supply voltage V_{dd} , a gate terminal coupled, preferably connected, to an output of the differential amplifier circuit **42** to receive the control voltage V_{ctrl} , and a drain coupled, preferably connected, to an output node **48** where a regulated output voltage V_{reg} is generated for application to a load circuit (not shown here, see FIG. 5 reference **102**). The feedback circuit **46** is formed by a resistive voltage divider circuit including a first resistor $R1$ coupled, preferably connected, in series at a feedback node **52** to a second resistor $R2$. The first resistor $R1$ is coupled, preferably connected, to the output node **48** and the second resistor $R2$ is coupled, preferably connected, to the ground node. The feedback voltage V_{fb} is generated at the feedback node **52** which is coupled, preferably connected, to the non-inverting ($+$) input of the differential amplifier circuit **42**.

(10) The voltage regulator circuit **40** further includes a first switched capacitor circuit **54** coupled between the gate terminal of the MOSFET device **44** and ground. This first switched capacitor circuit **54** includes a first capacitor $C1$ coupled in series with a first switch $S1a$ and a first voltage limiting circuit $M1$ between the gate terminal of the MOSFET device **44** and ground. The first voltage limiting circuit $M1$ may, for example, be implemented by a MOSFET device that comprises an n-channel type device (nMOS transistor) having a drain terminal coupled, preferably connected, to the gate terminal of the MOSFET device **44**, a gate terminal biased by a bias voltage V_b (so that the nMOS transistor has a desired drain-to-source resistance), and a source terminal coupled, preferably connected, to a first terminal of the first switch $S1a$. A second terminal of the first switch $S1a$ is coupled, preferably connected, to a first terminal of the first capacitor $C1$. A second terminal of the first capacitor $C1$ is coupled, preferably connected, to ground. The first switched capacitor circuit **54** further includes a second switch $S1b$ coupled, preferably connected, in parallel with the first capacitor $C1$.

(11) The actuation (closing) and deactuation (opening) of the first and second switches $S1a$ and $S1b$ are inversely controlled in response to the logic state of a first control signal $Ctrl1$. Specifically, first switch $S1a$ is open and second switch $S1b$ is closed in response to a first logic state (for example, deassertion) of signal $Ctrl1$, and conversely first switch $S1a$ is closed and second switch $S1b$ is open in response to a second logic state (for example, assertion) of signal $Ctrl1$. The first and second switches $S1a$ and $S1b$ may, for example, be implemented by MOSFET devices. A control terminal of the second switch $S1b$ is driven by control signal $Ctrl1$ and a control terminal of the first switch $S1a$ is driven by a logical inverse of the control signal $Ctrl1$ generated by an inverter circuit. When control signal $Ctrl1$ is in the first logic state, the second switch $S1b$ is closed (with the first switch $S1a$ open), and the first capacitor $C1$ is discharged to ground and the amplifier **42** sets the control voltage V_{ctrl} for the MOSFET device **44** in response to the comparison of the voltages V_{ref} and V_{fb} . When control signal $Ctrl1$ is in the second logic state, the first switch $S1a$ is closed (with the second switch $S1b$ open), the first capacitor $C1$ is coupled to the gate terminal of the MOSFET device **44** through the first voltage limiting circuit $M1$ and a first current $i1$ flows discharging the gate terminal of the MOSFET device **44** and charging the first capacitor $C1$. There is a corresponding decrease in the control voltage V_{ctrl} causing the MOSFET device **44** to turn on harder and source more output current i_{out} to the output node **48** than would otherwise be provided in response to the comparison of the voltages V_{ref} and V_{fb} performed by

amplifier **42** in setting the level of the control voltage V_{ctrl} . The total amount of charge drawn by capacitor **C1** is equal to $C1(V_b - V_{th})$, wherein V_{th} is the threshold voltage of the nMOS transistor forming the first voltage limiting circuit **M1**.

(12) The voltage regulator circuit **40** further includes a second switched capacitor circuit **56** coupled between the gate terminal of the MOSFET device **44** and ground. This second switched capacitor circuit **56** includes a second capacitor **C2** coupled in series with a third switch **S2a** and a second voltage limiting circuit **M2** between the gate terminal of the MOSFET device **44** and ground. The second voltage limiting circuit **M2** may, for example, be implemented by a MOSFET device that comprises an n-channel type device (nMOS transistor) having a drain terminal coupled, preferably connected, to the gate terminal of the MOSFET device **44**, a gate terminal biased by the bias voltage V_b (so that the nMOS transistor has a desired drain-to-source resistance), and a source terminal coupled, preferably connected, to a first terminal of the third switch **S2a**. A second terminal of the third switch **S2a** is coupled, preferably connected, to a first terminal of the second capacitor **C2**. A second terminal of the second capacitor **C2** is coupled, preferably connected, to ground. The second switched capacitor circuit **56** further includes a fourth switch **S2b** coupled, preferably connected, in parallel with the second capacitor **C2**.

(13) The actuation (closing) and deactuation (opening) of the third and fourth switches **S2a** and **S2b** are inversely controlled in response to the logic state of a second control signal $Ctrl2$. Specifically, third switch **S2a** is open and fourth switch **S2b** is closed in response to a first logic state (for example, deassertion) of signal $Ctrl2$, and conversely third switch **S2a** is closed and fourth switch **S2b** is open in response to a second logic state (for example, assertion) of signal $Ctrl2$. The third and fourth switches **S2a** and **S2b** may, for example, be implemented by MOSFET devices. A control terminal of the fourth switch **S2b** is driven by control signal $Ctrl2$ and a control terminal of the third switch **S2a** is driven by a logical inverse of the control signal $Ctrl2$ generated by an inverter circuit. When control signal $Ctrl2$ is in the first logic state, the fourth switch **S2b** is closed (with the third switch **S2a** open), and the second capacitor **C2** is discharged to ground, and the amplifier **42** sets the control voltage V_{ctrl} for the MOSFET device **44** in response to the comparison of the voltages V_{ref} and V_{fb} . When control signal $Ctrl2$ is in the second logic state, the third switch **S2a** is closed (with the fourth switch **S2b** open), the second capacitor **C2** is coupled to the gate terminal of the MOSFET device **44** through the second voltage limiting circuit **M2** and a second current $i2$ flows discharging the gate terminal of the MOSFET device **44** and charging the second capacitor **C2**. There is a corresponding decrease in the control voltage V_{ctrl} causing the MOSFET device **44** to turn on harder and source more output current i_{out} to the output node **48** than would otherwise be provided in response to the comparison of the voltages V_{ref} and V_{fb} performed by amplifier **42** in setting the level of the control voltage V_{ctrl} .

(14) The capacitances of the first and second capacitors **C1** and **C2** may be different. In particular, the capacitances may be sized in accordance with operational needs so that the first and second switched capacitor circuits **54** and **56** impose different degrees of decrease in the control voltage V_{ctrl} responsive to the first and second control signals $Ctrl1$ and $Ctrl2$ being in the second logic state. An example of this is described in more detail below in connection with FIG. 5.

(15) In an embodiment, only one switched capacitor circuit **54'**, instead of both the first and second switched capacitor circuits **54** and **56**, need be included in the regulator **40** as shown in FIG. 3. Switch **S1a** and **S1b** actuation in this switched capacitor circuit **54'** is controlled by control signal $Ctrl$.

(16) In an alternate embodiment, as shown in FIG. 4, the first and second voltage limiting circuits **M1** and **M2** may be omitted from the regulator **40**.

(17) Although illustrated in the context of an LDO type voltage regulator, the switched capacitor circuit **54'** (FIG. 3), or both the first and second switched capacitor circuits **54** and **56** (FIGS. 2 and 4), may be utilized at the gate terminal of the output MOSFET transistor for other types of voltage regulator circuits.

(18) Reference is now made to FIG. 5 which shows a block diagram of a voltage supply circuit **100** including the voltage regulator **40** (like that shown in FIGS. 2 and 4, for example) coupled to a load circuit **102** formed by a voltage charge pump (CP) circuit. The CP circuit includes a multi-stage (i.e., one or more stages) capacitor circuit **102a** having a voltage input node configured to receive the regulated output voltage V_{reg} from the voltage regulator **40** and a voltage output node **104** configured to generate a charge pump output voltage V_{cp} which is higher than the regulated output voltage V_{reg} . The multi-stage capacitor circuit **102a** further includes a control input configured to receive a clock signal CLK. An example of such a multi-stage capacitor circuit **102a**, without limitation or restriction, is a Dickson-type circuit well known to those skilled in the art where each stage includes two capacitors driven by opposite phases of the clock signal CLK corresponding to a charge input phase and a charge transfer phase. Another example of such a multi-stage capacitor circuit **102a**, without limitation or restriction, is voltage doubler-type circuit also well known to those skilled in the art where opposite phases of the clock signal CLK drive switching operations for a charge input phase where a flying capacitor is charged and a charge transfer phase where charge from the flying capacitor is transferred to an output capacitor.

(19) The clock signal CLK is generated by an oscillator circuit **102b**. It will be understood that the clock signal CLK generated by the oscillator circuit **102b** may be processed to generate multiple clock phases as needed to control the multi-stage capacitor circuit **102a**. The oscillator circuit **102b** is selectively enabled for operation in response to an enable signal En that is generated by a regulator circuit **102c**. The regulator circuit **102c** is coupled through a feedback circuit **106** to sense the charge pump output voltage V_{cp} at output node **104**. The feedback circuit **106** is formed by a resistive voltage divider circuit including a first resistor $R1_{out}$ coupled, preferably connected, in series at a feedback node to a second resistor $R2_{out}$. The first resistor $R1_{out}$ is coupled, preferably connected, to the output node **104** and the second resistor $R2_{out}$ is coupled, preferably connected, to the ground node. The sensed output voltage is generated at the feedback node and applied to the regulator circuit **102c**. If the regulator circuit detects that the charge pump output voltage V_{cp} is above the threshold level, the enable signal En is driven to a first logic state which controls the oscillator circuit **102b** to turn off and terminate output of the clock signal CLK. Conversely, if the regulator circuit detects through the sensed voltage that the charge pump output voltage V_{cp} is below some threshold level, the enable signal En is driven to a second logic state which controls the oscillator circuit **102b** to turn on and generate output of the clock signal CLK. Responsive to the oscillation of the clock signal CLK, the multi-stage capacitor circuit **102a** boosts the regulated voltage V_{reg} to increase the level of the charge pump voltage V_{cp} .

(20) The enable signal En of the charge pump circuit is applied to the voltage regulator **40** as the first control signal $Ctrl1$. When the enable signal En (first control signal $Ctrl1$) is in the first logic state corresponding to turning off the oscillator circuit **102b**, the second switch $S1b$ of the first switched capacitor circuit **54** in the voltage regulator **40** is closed (with the first switch $S1a$ open), and the first capacitor $C1$ is discharged to ground. However, when the enable signal En (first control signal $Ctrl1$) is the second logic state corresponding to turning on the oscillator circuit **102b**, the first switch $S1a$ of the first switched capacitor circuit **54** in the voltage regulator **40** is closed (with the second switch $S1b$ open), and the first current $i1$ flows from the gate terminal of MOSFET device **44** to charge the first capacitor $C1$. As a result, there is a drop in the gate voltage and additional current is sourced by MOSFET **44** to the regulator output to hold the regulator output voltage V_{reg} in response to the current spike to the load **102** which can occur during startup of the charge pump circuit.

(21) The clock signal CLK is applied to the voltage regulator **40** as the second control signal $Ctrl2$. When the clock signal CLK (second control signal $Ctrl2$) is the first logic state, the fourth switch $S2b$ of the second switched capacitor circuit **56** in the voltage regulator **40** is closed (with the third switch $S2a$ open), and the second capacitor $C2$ is discharged to ground. However, when the clock signal CLK (second control signal $Ctrl2$) is the second logic state, the third switch $S2a$ of the

second switched capacitor circuit **56** in the voltage regulator **40** is closed (with the fourth switch **S2b** open), and the second current i_2 flows from the gate terminal of MOSFET device **44** to charge the second capacitor **C2**. As a result, there is a drop in the gate voltage in response to the second logic state (phase) of the clock signal CLK and additional current is sourced by MOSFET **44** to the regulator output to hold the regulator output voltage V_{reg} in response to the current spike to the load **102** which can occur in response to switching operation of the charge pump circuit.

(22) With reference once again to the issue of choosing capacitance values for the first and second capacitors **C1** and **C2**, it will be noted that the current demand of the load **102** at startup of the charge pump circuit is likely to be greater than the current demand of the load at each charge input phase. Because of this, it would be advantageous to choose a capacitance for the first capacitor **C1** that is larger than the capacitance of the second capacitor **C2**. In view of the larger capacitance for capacitor **C1**, there will be a correspondingly larger drop in the voltage at the gate terminal of MOSFET device **44** causing the MOSFET device **44** to source a relatively larger difference in output current i_{out} to the load to compensate for the transient spike in current demand.

(23) In any case, it will be noted that the capacitances of the first and second capacitors **C1** and **C2** will be substantially smaller than the capacitance of the necessary filtering capacitor of the prior art solution. Such capacitors, with small capacitances, have a negligible impact on occupied surface area of the integrated circuit.

(24) It will further be noted that the solution described herein for use of one or more switched capacitor circuits **54'**, **54**, **56**, obviates the need to increase quiescent current and thus provided for improved power consumption which is advantageous in battery powered and wirelessly powered circuit applications.

(25) Reference is now made to FIG. **6** which illustrates a memory device **200** including a memory array **202**, a row decoder circuit **204**, a column decoder circuit **206** and a programming circuit **208**. The memory array **202** is of the type which requires for one or more of the write operation or the erase operation the application of a relatively high control voltage to the memory cells within the array. For example, this may comprise, in the context of a resistive memory of the phase change type, the application of a high voltage when instigating a phase change to program a certain logic state in a cell. The memory device **200** may further include the voltage supply circuit **100** to provide the charge pump output voltage V_{cp} to the programming circuit **108**.

(26) With reference to FIG. **7**, the memory device **200** may comprise an embedded memory within an integrated system on chip (SoC) device **210** including a digital circuit **212** (such as a microcontroller) and analog circuits **214**. The SoC device **210** may, for example, be a circuit component of a consumer electronic device **220** such as a smart phone, tablet computer, personal computer, television, battery charging circuit, communications devices, etc. More generally, the consumer electronic device **220** may comprise a consumer device powered from a battery or powered wirelessly.

(27) While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are considered illustrative or exemplary and not restrictive; the invention is not limited to the disclosed embodiments. Other variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims.

Claims

1. A circuit, comprising: a voltage regulator circuit configured to receive a reference voltage and generate a regulated output voltage, said voltage regulator circuit including a field effect transistor (FET) device having a gate terminal configured to receive a control voltage and a drain terminal configured to source an output current for generating the regulated output voltage; and a charge pump circuit having an input configured to receive the regulated output voltage, an output

configured to generate a pumped voltage, and a clock input configured to receive a clock signal, wherein pumping operation of said charge pump circuit is driven by opposite first and second logic states of the clock signal; wherein said voltage regulator circuit further includes a switched capacitor circuit coupled to the gate terminal and configured to selectively charge a capacitor with a charging current sunk from the gate terminal of the FET device in response to the first logic state of the clock signal and selectively discharge the capacitor in response to the second logic state of the clock signal.

2. The circuit of claim 1, wherein the charge pump circuit comprises a multi-stage capacitor circuit.

3. The circuit of claim 2, wherein a stage of the multi-stage capacitor circuit is implemented as a Dickson circuit.

4. The circuit of claim 2, wherein a stage of the multi-stage capacitor circuit is implemented as a voltage doubler circuit.

5. The circuit of claim 1, wherein the switched capacitor circuit further comprises a voltage limiting circuit coupled in series with the capacitor between the gate terminal and a ground.

6. The circuit of claim 5, wherein the voltage limiting circuit comprises a FET device having a gate terminal biased by a bias voltage.

7. The circuit of claim 1, wherein the voltage regulator circuit is a low drop out (LDO) type circuit.

8. The circuit of claim 1, wherein the switched capacitor circuit further comprises: a first transistor switch coupled in series with the capacitor between in the gate terminal of the MOSFET device and ground, said first transistor switch actuated to a closed state in response to the first logic state of the clock signal and actuated to an open state in response to the second logic state of the clock signal; and a second transistor switch coupled in parallel with the capacitor, said second transistor switch actuated to an open state in response to the first logic state of the clock signal and actuated to a closed state in response to the second logic state of the clock signal.
